Lab 4 – Defense Notes: Observers, Kalman Filter and Discrete Position Control

1. Model

Q: Why did the Lab 1 model have to be tuned?
Its $R_a = 72.7\,\Omega$ gave 0.084 A and 107 rpm at 12 V instead of 0.2 A and 200 rpm. R_a was set from the stall current ($12/5.5 = 2.18\,\Omega$), K_v and β from the two steady-state conditions of the guide, and J, L_a by least squares on the 4, 6, 8 V steps.

Q: Why does the model still overestimate the speed at low voltage?
Dry friction: a dead zone of ≈ 0.8 V that a linear model cannot represent.

Q: Why ZOH and $T_s = 10$ ms?
The Pico holds the PWM constant between samples, which is exactly a zero-order hold. $z = e^{sT_s}$: electrical mode ($\tau_e = 0.25$ ms) $\to z \approx 0$; mechanical mode (48 ms) $\to z = 0.812$.

2. Luenberger

Q: Is the system observable measuring only ω?
$\mathcal{O} = [C; CA] = [0\ 1;\ 0.8233\ 0.8166]$, $\det = -0.82 \neq 0 \Rightarrow$ rank 2.

Q: What does "predictor" mean?
$\hat{x}_k$ uses y_{k-1} and u_{k-1}: $\hat{x}_k = (A - LC)\hat{x}_{k-1} + Ly_{k-1} + Bu_{k-1}$. The error obeys $e_k = (A - LC)e_{k-1}$.

Q: Why poles $\{0, 0.2\}$ and not $\{0.3, 0.4\}$?
The plant's electrical pole is already at $z \approx 0$. Moving it to 0.3 makes the observer *slower* than the plant, and the model error accumulates in the current (-3.5 A estimates). Keep fast modes at $z \approx 0$ and only speed up the mechanical one.

3. Kalman

Q: Where do Q and R come from?
$R = 35\,(\text{rad/s})^2$: variance of the measured speed in steady state (26–45). $Q = \text{diag}(10^{-4}, 100)$: 10 mA and 10 rad/s of model uncertainty. $P_0 = \text{diag}(0.1, 100)$: larger initial doubt on the unmeasured current.

Q: What do prediction and correction do?
Prediction: the model advances the state and the uncertainty grows ($P^- = APA^T + Q$). Correction: the measurement reduces it ($P^+ = (I - KC)P^-$) with $K = P^-C^T/(CP^-C^T + R)$.

Q: What does $\sqrt{P_{11}^2 + P_{22}^2}$ show? Why doesn't it go to 0?
It shows convergence: $100 \to 28.93 \to 27.06 \to 26.99$, constant from the 3rd sample. With constant A, Q, R, P follows the Riccati equation to a fixed value independent of the data; it is not zero because model and sensor always have error. To make it vary, Q or R would have to change in time (adaptive filter).

Q: Why is Kalman better than Luenberger?
7.8 vs 34.8 rad/s RMS under load. Its gain comes from Q, R (not hand-picked poles) and its speed gain 0.771 weights the measurement more, filtering the 9.8 rad/s encoder quantization.

Q: Why is the estimated speed not zero with the motor braked?
Neither model has the load torque: the observer expects the motor to turn whenever voltage is applied. Fix: add the load torque as a third state.

4. Discrete position control (additional)

Q: How was it designed in discrete time?
Plant $G(z) = 5.37\times10^{-4}(z+1.029)/((z-1)(z-0.812))$ (ZOH; the electrical pole at $z \approx 0$ cancels with a zero at -0.0012). Controller $C(z) = (q_0z^2 + q_1z + q_2)/((z-1)(z+g))$. Matching $(z-1)(z+g)D + N(q_0z^2 + q_1z + q_2)$ with the desired polynomial gives 4 linear equations.

Q: Where do the poles come from?
$\zeta = 0.7$, $t_s = 0.5$ s $\Rightarrow s = -8 \pm j8.16 \Rightarrow z = e^{sT_s} = 0.92 \pm 0.075j$; two extra poles at 0.6.

Q: Why not cancel the zero at -1.029?
It is a sampling zero outside the unit circle; cancelling it would create an unstable hidden mode.

Q: Why the prefilter and the anti-windup?
$q_0 = 140$ V/rad: a 1 rad step asks > 100 V. Saturated at 6 V the voltage swung $+6 \to -6$ V. $F(z) = ((1-\alpha)z/(z-\alpha))^2$, $\alpha = 0.93$, smooths the reference without moving the closed-loop poles. Storing the saturated u in the difference equation stops the integral from winding up.

Q: Did the design hold on the robot?
27 steps: $t_s = 0.74$–0.85 s (design 0.65–0.75), overshoot $< 1\%$, mean final error 1.8 mrad. The extra 0.1 s and the few mrad are dry friction.

Q: Why does the controller hold ≈ 2 V at rest?
External torque toward mid-travel (cables). The integral cancels it; the observers read it as current.

5. Implementation

Q: What runs where?
Pico (10 ms loop): encoder, both observers, PID, PWM, limits, stall cut, shoulder hold (PD in a separate timer). PC: design in Python (headers), web interface only sends commands and plots.

Q: Why no matrix inversion in Kalman?
$C = [0\ 1] \Rightarrow$ the innovation is scalar: $K = P^-_{:,2}/(P^-_{22} + R)$.